

TrES-5b

R-0901

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_250807 · created 2026-07-29T03:45:37+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460894.9631 +/- 0.0043 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.055
Transit depth (Rp/Rs)^2	3.33 +/- 1.99 [%]
Orbital Inclination (inc)	82.35 +/- 2.62
Airmass coefficient 1 (a1)	0.584 +/- 0.079
Airmass coefficient 2 (a2)	0.4 +/- 0.13
Scatter in the residuals of the lightcurve fit is	14.6 %
Best Comparison Star	#2 - [304, 422]
Optimal Aperture	3.34
Optimal Annulus	9.36
Transit Duration (day)	0.053 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.055 vs 0.142 (deviation 0.44 σ)
Duration fit vs published	0.053 d vs 0.0758 d (deviation 0.59 σ)
Mid-time O-C	9.9 min
Depth SNR	2.0
Residual scatter	14.6 %

Light curve

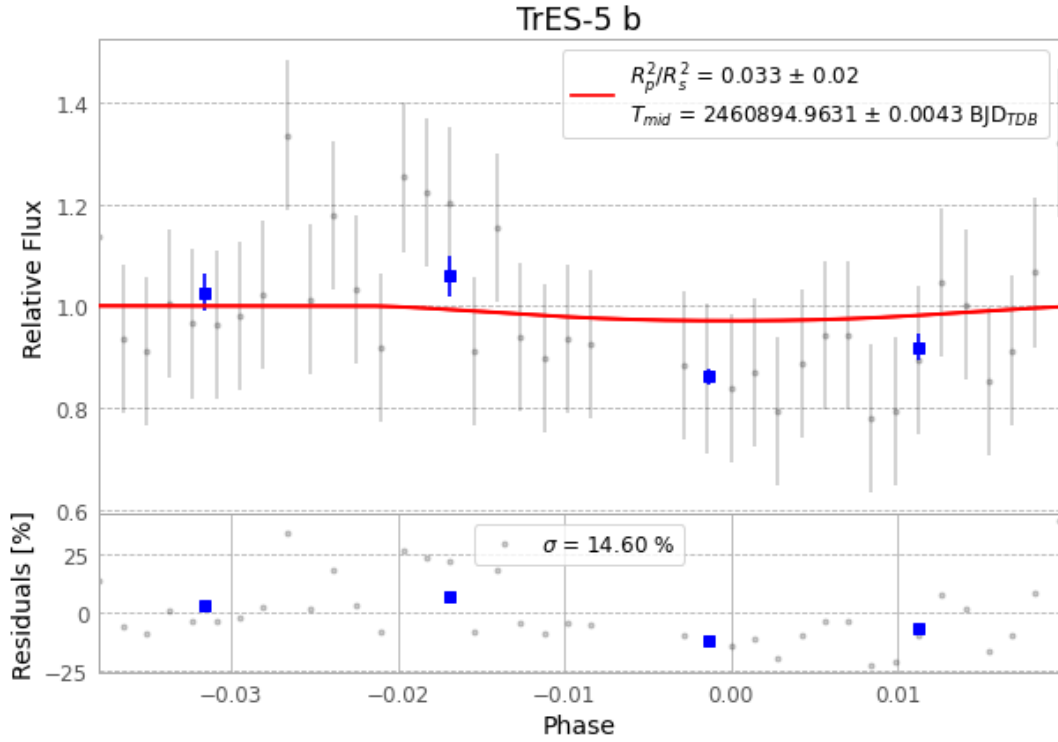


Figure 1 — FinalLightCurve_TrES-5 b_2025-08-07.png (SHA-256 7de9967ddc577647...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [319, 284], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b4241c13becece1d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

42 FITS frames (28 MB), TRES-5250807094215.FITS → TRES-5250807114515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-250807.log (938f0526bd43...), FinalLightCurve_TrES-5 b_2025-08-07.png (7de9967ddc57...), FinalLightCurve_TrES-5 b_2025-08-07.csv (b2db47145314...)

Compute resources

Wall time 205.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 03:45Z.